

Yewei Huang, Ph.D.

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RESEARCH INTEREST

My main research interest is the challenges associated with **robot perception, navigation and decision-making** problems, particularly in the context of **multi-robot systems**. By deploying these algorithms in real-world environments on **physical robot systems** with **limited communication bandwidth**, I aim to enable fully autonomous robots to support the long-term maintenance of the underwater infrastructures and effectively monitor the complex offshore hydrodynamic and environmental conditions.

I'm interested in marine robot, multi-robot systems, SLAM, exploration, inter-robot loop closure, object-based data association, factor graph optimization, distributed multi-robot SLAM, planning under uncertainty, Gaussian process motion planning, expectation-maximization, learning-based decision making, multi-sensor fusion.

EDUCATION

<i>Doctor of Philosophy</i> , Mechanical Engineering Stevens Institute of Technology, NJ	08/2019 - 05/2025
<i>Master of Science</i> , Surveying Engineering Tongji University, Shanghai, China	09/2016 - 03/2019
<i>Bachelor of Science</i> , Geographic Information System Tongji University, Shanghai, China	09/2012 - 06/2016

AWARDS AND HONORS

- Academic Year 2019-2020 Stevens Provost Doctoral Fellowship
- Academic Year 2024-2025 Stevens Excellence Doctoral Fellowship
- Maryland Robotics Center: Future Leaders in Robotics and AI of 2025
- Academic Year 2024-2025 Paul Kaplan Award for Distinguished Doctoral Work upon Graduation
- Honorable Mention for the inaugural IEEE RAS Women in Engineering Best PhD Award in Robotics and Automation (2026)

RESEARCH EXPERIENCE

<i>Post-doctoral Research Fellow</i> , Dartmouth Reality and Robotics Lab Dartmouth College, Hanover, NH Advisor: Dr. Alberto Quattrini Li	07/2025 - present
• Developing 3D sonar SLAM and sonar-camera calibration for underwater robots. • Developing marine exploration and coastal path planning under uncertainty through projects supported by the National Science Foundation (NSF).	
<i>Summer Research Internship</i> , Vision and Robotics Laboratory SRI International, Princeton, NJ Advisor: Dr. Han-Pang Chiu	05/2024 - 08/2024
• Developed SGM-SLAM, a distributed multi-robot SLAM framework using object-level scene graph matching for efficient communication and shared scene understanding.	

- Evaluated scene graph mapping with a quadruped robot in indoor and outdoor environments as part of the SUWAC project, which was later licensed to Avsr AI.

Graduate Student Research Assistant, Robust Field Autonomy Laboratory

Stevens Institute of Technology, Hoboken, NJ

08/2019 - 05/2025

Advisor: Dr. Brendan Englot

- Developed [DiSCo-SLAM](#) and [DRACo-SLAM2](#) for distributed multi-robot LiDAR and sonar mapping under limited communication, evaluated on data from ground and underwater robots.
- Developed [multi-robot exploration under localization uncertainty](#) and [Gaussian process motion planning](#) accounting for seafloor terrain and ocean currents.
- Developed [geo-referenced structure-from-motion](#) with geographic constraints incorporated into bundle adjustment.

Contributed to projects supported by NSF, the Office of Naval Research (ONR), DARPA, and IARPA.

PUBLICATIONS

1. Y. Li, **Y. Huang**, W. Gaozhang, A. Quattrini Li, B. Englot, and Y. Liu “[Variable-Resolution Virtual Maps for Autonomous Exploration with Unmanned Surface Vehicles \(USVs\)](#)” *Submitted, under review, 2026*.
2. J. McConnell, **Y. Huang**, T. Morris, J. Doughty, and D. Moynihan “[Multi-session SLAM for Imaging Sonar Equipped Underwater Vehicles Using Semantic Scene Graphs](#)” *IEEE Robotics and Automation Letters (RA-L)*, 2026.
3. C. Burgul, **Y. Huang**, M. Chatzisprou, I. Rekleitis, A. Quattrini Li, and M. Xanthidis “[Underwater dense mapping with the first compact 3d sonar](#)” *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.
4. C. Burgul, X. Zhao, **Y. Huang**, M. Chatzisprou, A. Quattrini Li, and I. Rekleitis “SVIn+: Multi-Modal Framework for Robust Underwater State Estimation using 3D Sonar, Visual-Inertial, Water-Pressure, and DVL” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026. (*Accepted, to appear*).
5. Y. Jiang, **Y. Huang**, W. Cao, M. Jeong, A. Quattrini Li, L. Zhao, M. Chen, and D. Balkcom “[Manipulider: A Multi-Engine Buoyancy-Controlled Robot for Thrusterless Underwater Gliding and Manipulation](#)” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026. (*Accepted, to appear*).
6. S. Ma, Y. Wang, **Y. Huang**, R. Bucknall, and Y. Liu “Active Tracking of Marine Pollution Sources: An Uncertainty-Aware Categorical Bayesian Framework for Unmanned Surface Vehicles” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026. (*Accepted, to appear*).
7. **Y. Huang**, T. Shan, A. Rajvanshi, N. Chowdhury, Y. Li, B. Englot, and H.-P. Chiu “[SGM-SLAM: Scene Graph Matching for Data-Efficient Distributed SLAM](#)” *IEEE International Symposium on Safety, Security, and Rescue Robotics (SSRR)*, 2026. (*Accepted, to appear*).
8. **Y. Huang**, J. McConnell, X. Lin, and B. Englot “[DRACo-SLAM2: Distributed Robust Acoustic Communication-efficient SLAM for Imaging Sonar Equipped Underwater Robot Teams with Object Graph Matching](#)” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.
9. Y. Li, **Y. Huang**, W. Gaozhang, and Y. Liu “[Autonomous Exploration with Virtual Map using Unmanned Surface Vehicles](#)” *IEEE International Conference on Automation and Computing (ICAC)*, 2025.
10. Y. Li, **Y. Huang**, B. Gaudel, H. Jafarnejadsani, and B. Englot “[CVD-SfM: A Cross-View Deep Front-end Structure-from-Motion System for Sparse Localization in Multi-Altitude Scenes](#)” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 10741-10748, 2025.

11. **Y. Huang**, X. Lin, and B. Englot “[Multi-Robot Autonomous Exploration and Mapping Under Localization Uncertainty with Expectation-Maximization](#)” *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 7236–7242, 2024.
12. **Y. Huang**, X. Lin, M. Hernandez-Rocha, S. Narain, K. Pochiraju and B. Englot “[Mission-oriented Gaussian Process Motion Planning for UUVs over Complex Seafloor Terrain and Current Flows](#)” *IEEE Robotics and Automation Letters (RA-L)*, 9(2):1780-1787, 2024.
13. X. Lin, P. Szenher, **Y. Huang**, and B. Englot “[Distributional Reinforcement Learning Based Integrated Decision Making and Control for Autonomous Surface Vehicles](#)” *IEEE Robotics and Automation Letters (RA-L)*, 2024.
14. X. Lin, **Y. Huang**, F. Chen, and B. Englot “[Decentralized Multi-Robot Navigation for Autonomous Surface Vehicles with Distributional Reinforcement Learning](#)” *IEEE International Conference on Robotics and Automation (ICRA)*, 2024.
15. K. Doherty, A. Papalia, **Y. Huang**, D. Rosen, B. Englot, and J. Leonard “[MAC: Maximizing Algebraic Connectivity for Graph Sparsification](#)” *arXiv e-prints*, arXiv-2403, 2024.
16. X. Lin, **Y. Huang**, D. Sun, T.-Y. Lin, B. Englot, R. M. Eustice, and M. Ghaffari “[A Robust Keyframe-Based Visual SLAM for RGB-D Cameras in Challenging Scenarios](#)” *IEEE Access*, 11, 97239–97249, 2023.
17. J. Wang, F. Chen, **Y. Huang**, J. McConnell, T. Shan, and B. Englot “[Virtual maps for autonomous exploration of cluttered underwater environments](#)” *IEEE Journal of Oceanic Engineering*, 47(4):916–935, 2022.
18. J. McConnell, **Y. Huang**, P. Szenher, I. Collado-Gonzalez and B. Englot “[DRACo-SLAM: Distributed Robust Acoustic Communication-efficient SLAM for Imaging Sonar Equipped Underwater Robot Teams](#)” *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 8457–8464, 2022.
19. F. Chen, P. Szenher, **Y. Huang**, J. Wang, T. Shan, B. Shi and B. Englot “[Zero-shot reinforcement learning on graphs for autonomous exploration under uncertainty](#)” *2021 IEEE International Conference on Robotics and Automation (ICRA)*, 5193–5199, 2021.
20. **Y. Huang**, T. Shan, F. Chen and B. Englot “[DiSCo-SLAM: Distributed scan context-enabled multi-robot lidar slam with two-stage global-local graph optimization](#)” *IEEE Robotics and Automation Letters (RA-L)*, 7(2):1150–1157, 2021.
21. F. Chen, J. Martin, **Y. Huang**, J. Wang, and B. Englot “[Autonomous exploration under uncertainty via deep reinforcement learning on graphs](#)” *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 6140–6147, 2020.
22. J. Zhao, **Y. Huang**, X. He, S. Zhang, C. Ye, T. Feng, and L. Xiong “[Visual semantic landmark-based robust mapping and localization for autonomous indoor parking](#)” *Sensors*, 19(1):161, 2019.
23. **Y. Huang**, J. Zhao, X. He, S. Zhang, and T. Feng “[Vision-based semantic mapping and localization for autonomous indoor parking](#)” *Proceedings of the IEEE Intelligent Vehicles Symposium (IV)*, 636–641, 2018.
24. **Y. Huang**, H. Wang, K. Zhan, J. Zhao, P. Gui, and T. Feng “[Image-based localization for indoor environment using mobile phone](#)” *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, 40:211–215, 2015.

MENTORING

I have mentored the following undergraduate and master’s students with my doctoral advisor, Prof. Brendan Englot:

- **Thomas Roff**. “Optimization of Utility Function Weights for Active Autonomous Exploration and Mapping With Multi-robot Systems” *Master’s thesis*, Stevens Institute of Technology, 2024.

Y. Huang
p. 4

- **Sneha Banda, Jesse Knuckles.** “Modular Sensor Fusion Package for Localization” *Undergraduate Senior Design*, sponsored by Kearfott Guidance and Navigation, Stevens Institute of Technology, 2025.

I have mentored the following doctoral student in collaboration with Prof. Yuanchang Liu from UCL:

- **Ye Li.** “Autonomous Exploration with Virtual Map using Unmanned Surface Vehicles” *The 30th International Conference on Automation and Computing (ICAC)*, 2025.

I have mentored the following doctoral student with my postdoctoral advisor, Prof. Alberto Quattrini Li, and Prof. Ioannis Rekleitis from University of Delaware:

- **Chinmay Burgul.** “Underwater dense mapping with the first compact 3d sonar” *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.
“SVIn+: Multi-Modal Framework for Robust Underwater State Estimation using 3D Sonar, Visual-Inertial, Water-Pressure, and DVL” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.

I have mentored the following undergraduate student in collaboration with Prof. Weidi Wang from Umass Lowell:

- “A New Design of an Underwater Newton Subsea Gripper for Marine Trash Collection” *Undergraduate Capstone*, University of Massachusetts Lowell, 2026.

SERVICES

Editorial Board

- Associate Editor, IEEE International Conference on Robotics and Automation (ICRA), 2026, 2027.
- Session Chair, “Sensing and Calibration in Marine Environments,” and Session Co-Chair, “Controlling Marine Vehicles,” IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026.

Reviewer

- IEEE Transactions on Robotics (T-RO).
- International Journal of Robotics Research (IJRR).
- IEEE Transactions on Field Robotics (T-FR).
- IEEE Robotics and Automation Letters (RA-L).
- Journal of Field Robotics (JFR).
- Elsevier Applied Ocean Research.
- Robotics: Science and Systems (RSS).
- IEEE International Conference on Robotics and Automation (ICRA).
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).

TALKS

- 03/28/2025, Maryland Robotics Center (University of Maryland): Future Leaders in Robotics and AI Seminar
Data Efficient Localization and Mapping for Distributed Multi-Robot Teams in the Field
- 07/11/2025, University College London: Invited talk (virtual),
Distributed Robust Acoustic Communication-efficient SLAM for Imaging Sonar Equipped Underwater Robot Teams with Object Graph Matching

- 02/13/2026, School of Civil and Environmental, University of Connecticut: Environmental Engineering Seminar Talk
Reliable Navigation and Mapping in Uncertain Coastal Waters: Towards Trustworthy Marine Autonomy
- 04/27/2026, Stevens Institute of Technology: ME 656 Autonomous Navigation for Mobile Robots
Machine Learning for Visual SLAM from Scene Representation Perspective
- 05/08/2026, Dartmouth College: COSC 81/281 Principles of Robot Design and Programming
Machine Learning for Visual SLAM from Scene Representation Perspective
- 09/30/2026, Virginia Tech: AOE 4984 Robot Perception
Simultaneous Localization and Mapping (SLAM): From Robot Localization to Underwater Mapping

SKILLS

C++, Python, ROS, reinforcement learning, computer vision, Matlab, LaTeX, Git.